

Loss, Learning & Generalisation

ECMM422

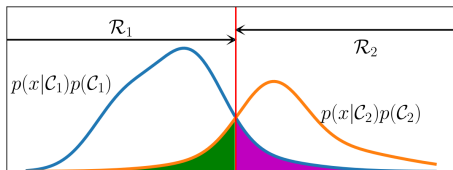
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Outline

- 1 Classification Loss
- 2 Unequal loss
 - Binary classification with unequal loss
- 3 Receiver Operating Characteristic
 - Convex hull
 - Comparing classifiers



Loss functions

The error functions for regression and classification were derived by maximising the likelihood:

$$E(w) = - \sum_{n=1}^N \log p(t_n | x_n, w)$$

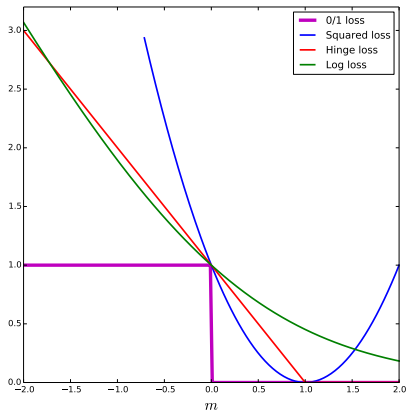
In general, for a *scoring classifier*, learn by minimising a regularised error

$$E_{\alpha}(w) = \underbrace{\sum_n^N L(t_n, \mathcal{F}(x_n; w))}_{\text{Data error}} + \underbrace{\alpha E_{\text{reg}}}_{\text{Regularisation}}$$

where $L : \mathbb{R} \times \mathbb{R} \mapsto \mathbb{R}$ is a *loss function* measuring how much we care about mismatches between the prediction and the data

Loss functions

- ▶ **Q:** How are loss functions different?
- ▶ **A:** They penalise large errors in different ways.
- ▶ **Aim:** characterize loss functions in terms of their robustness to extreme data.



Classification loss functions

- ▶ Define the targets to be $t_n \in \{-1, +1\}$
- ▶ Transform the score to a *decision* by classifying according to the *sign* of $\mathcal{F}(x_n; w)$
- ▶ i.e. the classification rule is: $G(x_n) = \text{sign}[\mathcal{F}(x_n; w)]$

Margin

$$m_n(w) = t_n \cdot \mathcal{F}(x_n; w)$$

m is the *margin*

$m > 0$ the target and the classifier agree (correct decision)

$m < 0$ the target and the classifier disagree (incorrect decision)

Classification loss functions

- ▶ Goal of the classification algorithm: produce positive margins as frequently as possible
- ▶ A loss criterion used for classification should penalize negative margins more heavily than positive ones since positive margin observations are already correctly classified

$$E(w) = \sum_n L(m_n(w))$$

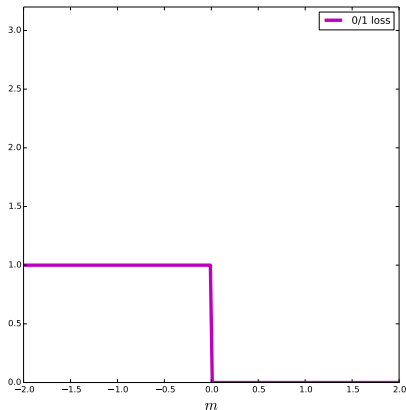
where $L(\cdot)$ is a *loss function*.

$m > 0$ loss function should be small because x_n is correctly classified

$m < 0$ loss function should be large because x_n is misclassified

$$L_{01}(m) = \begin{cases} 0 & \text{if } m \geq 0 \\ 1 & \text{if } m < 0 \end{cases}$$

- ▶ This loss counts the number of mistakes the classifier makes.
- ▶ Gradient is zero almost everywhere: no information on how to reduce error!



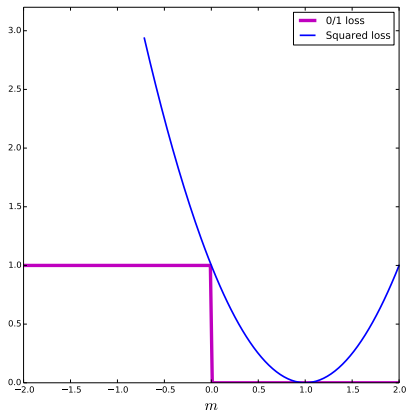
Squared loss

$$E_2(w) = (\mathcal{F}(x; w) - t)^2$$

Scale so that loss is 1 when $m = 0$ and loss is 0 when $m = 1$.
Thus

$$L_2(m) = (m - 1)^2$$

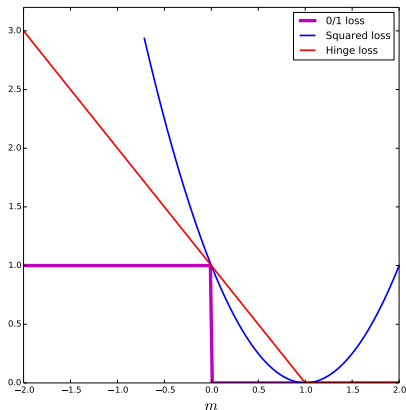
- ▶ Appropriate for regression, not classification.
- ▶ In classification setting: penalises features that are correctly classified $m > 1$



Hinge loss

$$\begin{aligned}L_{\text{hinge}} &= \max(0, 1 - t \cdot \mathcal{F}(x; w)) \\&= \max(0, 1 - m) \\&= [1 - m]_{+}\end{aligned}$$

- ▶ Used in large margin classifiers.
- ▶ No penalty to correctly classified features, but a linearly increasing penalty to patterns that fall in the margin $m \in (0, 1)$ and to incorrectly classified patterns outside the margin $m < 0$.



Log loss

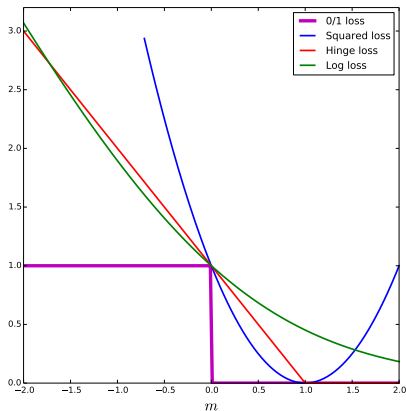
Score $\mathcal{F}(x; w) \in \mathbb{R}$ is transformed to a probability with the *logistic link function*

$$p(t | x; w) = \frac{1}{1 + e^{-\mathcal{F}(x)}}$$

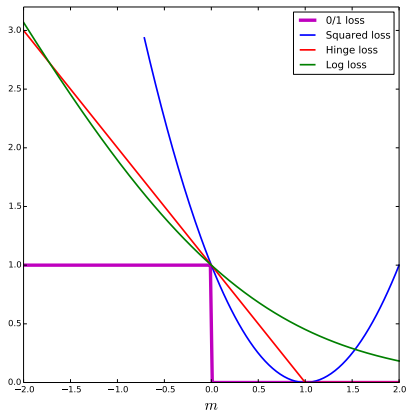
Then the loss can be written as

$$L_{\log} = \log(1 + e^{-m}) / \log(2)$$

Penalises incorrectly classified features more strongly as m becomes more negative



When robustness is a concern
(e.g. outliers are possible)
squared- error loss for regression
and exponential loss for
classification are not the best
criteria



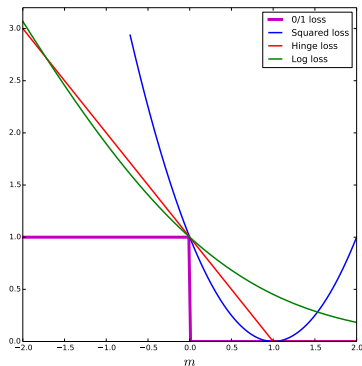
Loss functions

Surrogates of L_{01}

- ▶ All the loss functions are convex surrogates of L_{01}

Probability

- ▶ If the goal is to maximise $p(t' | x')$ for new data x' , then log-loss is correct function
- ▶ But with an unbounded loss, infinitely bad observations are possible



Bounded loss and convexity

- ▶ Reasonable that loss should be bounded and L_{01} is often used as a measure of success
- ▶ But bounded loss implies non-convexity and difficult to optimise functions. Convex approximations/surrogates are easier to work with

- ▶ Probability theory gives us a consistent mathematical framework for quantifying and manipulating uncertainty
- ▶ Decision theory (combined with probability theory), allows optimal decisions in situations involving uncertainty
 - ▶ Inference allows us to determine $p(x, t)$ but then we need to take a specific action based on our understanding of the values t is likely to take
 - ▶ E.g. given X-ray image (x is the pixel intensities), we predict t the presence of cancer: how to best choose the mapping between the values of t and the classes $\mathcal{C}_1 = \text{presence of cancer}$ and $\mathcal{C}_2 = \text{absence of cancer}$?
 - ▶ How to combine t with the notion that different types of errors have different associated costs?

Loss matrix

The consequences of misclassifications frequently are not equally costly, although the loss function assumes that they are.

Classification of tumours

		True class	
		$\mathcal{C}_1$, cancer	$\mathcal{C}_2$, benign
Predicted classification	$\mathcal{C}_1$, cancer	1	1.5
	$\mathcal{C}_2$, benign	10	0

Loss matrix λ_{ij} quantifies the penalty of assigning x to $\mathcal{C}_i$ when it belongs to $\mathcal{C}_j$

Often we count the cost of correct classification as zero, so $\lambda_{ii} = 0$ for all i .

Risk Average cost of making a classification.
Averaged over all classes.

Let's formalise the various notions that are needed:

- ▶ state of nature or **class** C_j
- ▶ **actions** $\alpha_1, \dots, \alpha_c$, a finite set of actions (i.e. predicted classes)
- ▶ **loss** $\lambda(\alpha_i|C_j)$ i.e. the cost that we pay for performing action α_i when the true state of nature is C_j
- ▶ **conditional risk** $R(\alpha_i|x) = \sum_{j=1}^c \lambda(\alpha_i|C_j)p(C_j|x)$, i.e. the expected loss
- ▶ **decision rule** $\alpha(x)$, a function that returns the action α_i to take for any observation x
- ▶ **risk** $R = \int R(\alpha(x)|x)p(x)dx$, i.e. the expected loss associated with the decision rule

- ▶ We want to find the decision rule that achieves minimal risk
- ▶ ...that is we want to choose $\alpha(x)$ so that $R(\alpha(x))$ is as small as possible for any x
- ▶ therefore we use the **Bayes decision rule**:
 - ▶ for all i compute the conditional risk
$$R(\alpha_i|x) = \sum_{j=1}^c \lambda(\alpha_i|C_j)p(C_j|x)$$
 - ▶ select α_i for which $R(\alpha_i|x)$ is minimum

Minimum Error Rate Classification

- ▶ In a classification problem we have c classes C_j
- ▶ The action α_i is to decide that the state of nature is C_i
- ▶ A typical loss function is the **zero-one loss**
 $\lambda(\alpha_i|C_j) = 0$ if $i = j$ and 1 otherwise
- ▶ ...that is we pay 0 if we are right and 1 for any error
- ▶ In this case the risk is equal to the average probability of error, since

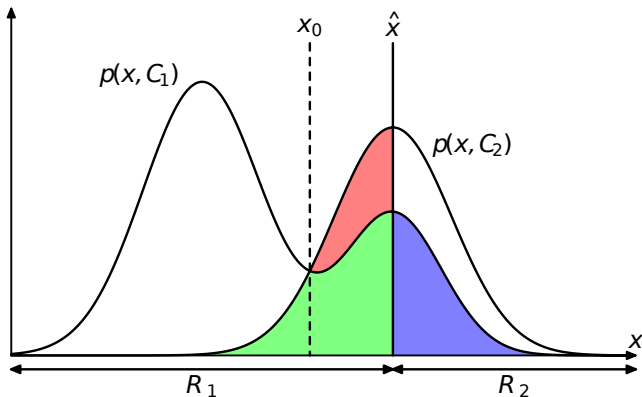
$$\begin{aligned} R(\alpha_i|x) &= \sum_{j=1}^c \lambda(\alpha_i|C_j)p(C_j|x) \\ &= \sum_{i \neq j} p(C_j|x) \\ &= 1 - p(C_i|x) \end{aligned}$$

In conclusion:

- ▶ To minimize the risk $\Rightarrow$ minimize the conditional risk
- ▶ To minimize the conditional risk $\Rightarrow$ minimize the average probability of error
- ▶ To minimize the average probability of error $\Rightarrow$ maximize the posterior $p(C_i|x)$
- ▶ Note that one can use the Bayes Theorem to express the posterior as $p(x|C_i)p(C_i)$

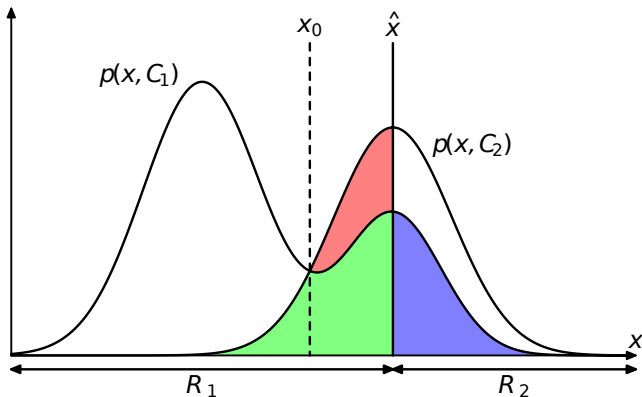
Minimum Error Rate Classification

- ▶ To visualize consider a 1 dimensional case
- ▶ R_1 is the **decision region** where the decision rule decides C_1 and R_2 for C_2
- ▶ The **decision boundary** or **decision surface** is $\hat{x}$



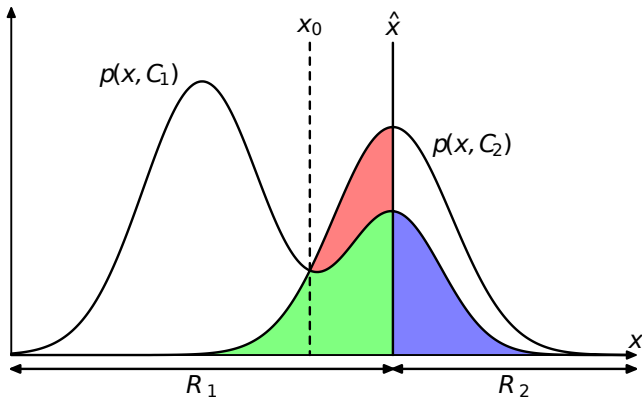
Minimum Error Rate Classification

- ▶ Errors are visualized by the green, blue and red regions
- ▶ Red+green=misclassification of points in C_2 as belonging to R_1 and hence to C_1
- ▶ Blue=misclassification of points in C_1 as belonging to R_2



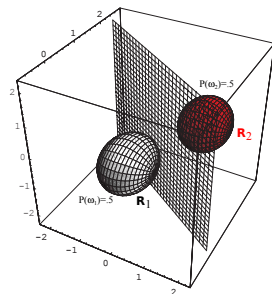
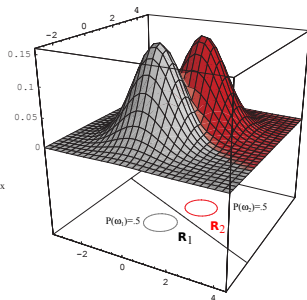
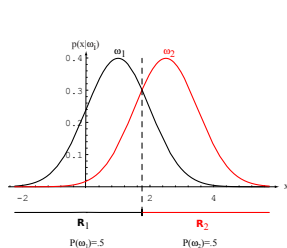
Minimum Error Rate Classification

- ▶ If we move $\hat{x}$ towards the optimal x_0 we can eliminate the red region
- ▶ x_0 is reached when $p(C_1|x) = p(C_2|x)$
- ▶ ..however we cannot reduce the sum green+blue
- ▶ The irreducible error is called the **Bayes error**



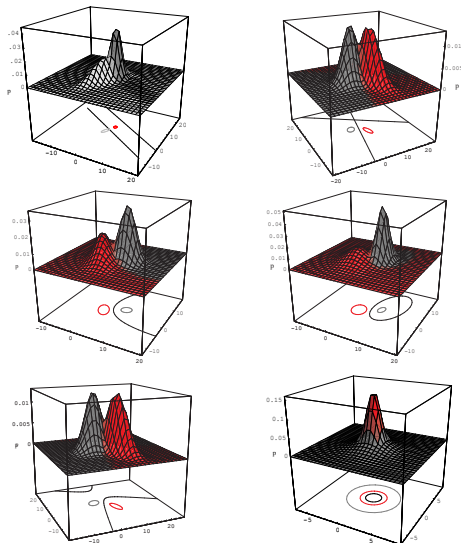
Bayes Decision Surface

What is the decision surface when $p(x|C)$ are Gaussians with identical variance?



Bayes Decision Surface

What is the decision surface when $p(x|C)$ are Gaussians with different covariance matrices?



Binary classification with unequal loss

For simplicity let: $\lambda_{ij} = \lambda(\alpha_i | C_j)$

Conditional risk of assigning x to C_1 or C_2

$$R(C_1 | x) = \lambda_{11}p(C_1 | x) + \lambda_{12}p(C_2 | x)$$

$$R(C_2 | x) = \lambda_{21}p(C_1 | x) + \lambda_{22}p(C_2 | x)$$

For minimum risk, assign to C_1 if $R(C_1 | x) < R(C_2 | x)$:

$$\lambda_{11}p(C_1 | x) + \lambda_{12}p(C_2 | x) < \lambda_{21}p(C_1 | x) + \lambda_{22}p(C_2 | x)$$

$$p(C_2 | x)(\lambda_{12} - \lambda_{22}) < p(C_1 | x)(\lambda_{21} - \lambda_{11})$$

applying Bayes rule:

$$p(x | C_2)p(C_2)(\lambda_{12} - \lambda_{22}) < p(x | C_1)p(C_1)(\lambda_{21} - \lambda_{11})$$

Likelihood ratio and decisions

There are a variety of ways of expressing the minimum-risk decision rule:
For minimum risk, assuming $\lambda_{21} - \lambda_{11} > 0$ assign to C_1 if

Likelihood ratio

$$\frac{p(x | C_1)}{p(x | C_2)} > \frac{p(C_2)(\lambda_{22} - \lambda_{12})}{p(C_1)(\lambda_{11} - \lambda_{21})}$$

Posterior ratio

$$\frac{p(x | C_1)p(C_1)}{p(x | C_2)p(C_2)} = \frac{p(C_1 | x)}{p(C_2 | x)} > \frac{\lambda_{22} - \lambda_{12}}{\lambda_{11} - \lambda_{21}}$$

- ▶ Note that the ratio rules are relative to a threshold that is independent of the observation x
- ▶ Note that often $\lambda_{11} = \lambda_{22} = 0$ and the ratio threshold becomes $\lambda_{12}/\lambda_{21}$

Binary classification

- ▶ With only two classes we often focus on a particular class of interest. (For example: classification of symptoms as a tumour rather than benign; detection that two aircraft are likely to collide rather than safely miss.)
- ▶ Usual to assign names to the elements of the confusion (rate) matrix; focusing on $\mathcal{C}_1$:

		True class	
		$\mathcal{C}_1$	$\mathcal{C}_2$
Classification	$\mathcal{C}_1$	True positive	False positive
	$\mathcal{C}_2$	False negative	True negative

- ▶ the meaning of the terms is interpreted as follows:
(1 prediction agrees with truth) (2 prediction):
 - ▶ so: (1 false) (2 positive) is interpreted as
(1 prediction does not agree with truth) (2 predicted positive)

- ▶ The true positive rate (TPR, also called **sensitivity**) is calculated as TP/P or $TP/(TP+FN)$
 - ▶ TPR is the probability that an actual positive will test positive
- ▶ The false positive rate (FPR) is calculated as FP/N or $FP/(FP+TN)$
- ▶ The true negative rate (TNR, also called **specificity**) is calculated as TN/N or $TN/(TN+FP)$
 - ▶ TNR is the probability that an actual negative will test negative
- ▶ The false negative rate (FNR) is calculated as FN/P or $FN/(TP+FN)$

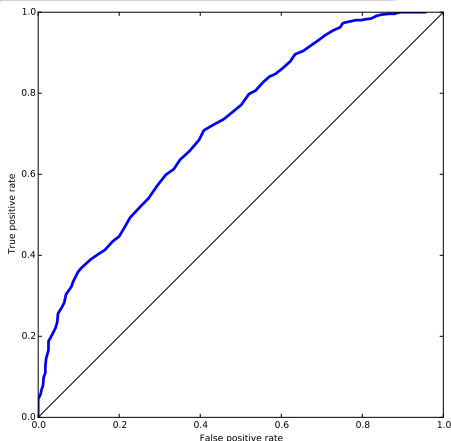
In the following we consider TPR and FPR

Receiver Operating Characteristic

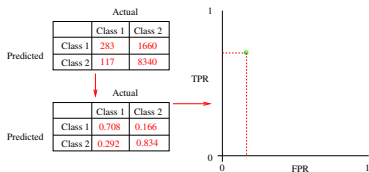
ROC curve

The ROC curve sweeps out the true positive rate versus the false positive rate for $0 \leq t \leq 1$.

- ▶ $0 \leq t \leq 1$ is equivalent to varying the ratio of misclassification costs from 0 to ∞
- ▶ The perfect classifier has ROC curve $(0,0) \rightarrow (0,1) \rightarrow (1,1)$.
- ▶ The diagonal line shows the performance of the classifier that allocates at random.

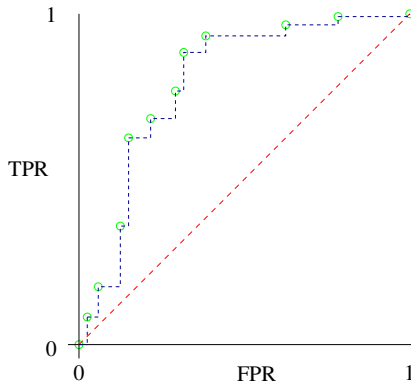


Constructing ROC curves

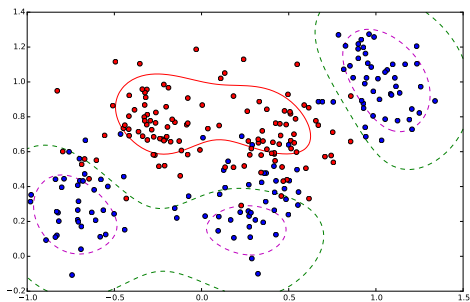
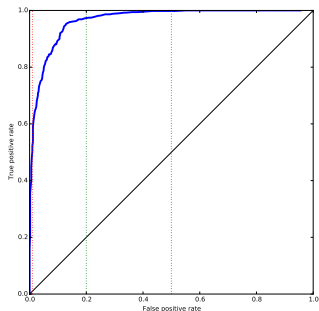
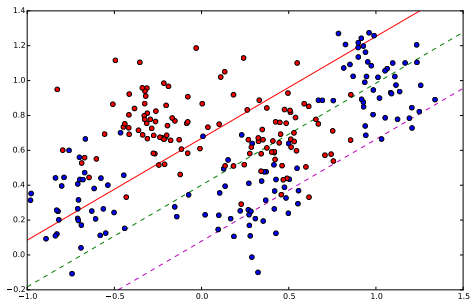
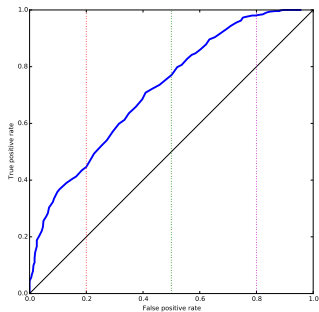


► For each threshold:

- 1 Evaluate confusion matrix
- 2 Plot TPR vs FPR



Data classification with a linear SVM and a RBF SVM

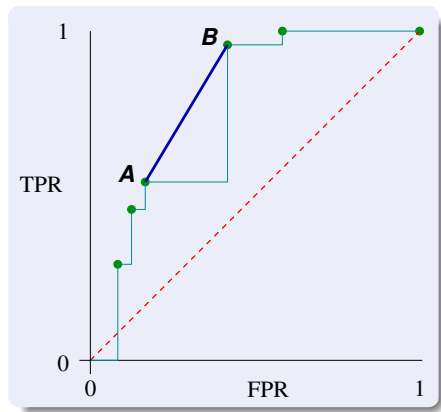


Convex hull

- ▶ Suppose classifier A has operating point (F_A, T_A) and classifier B has operating point (F_B, T_B)
- ▶ Then construct a new classifier by randomly choosing A for a proportion ω of classifications and B for the remaining proportion $(1 - \omega)$.
- ▶ The new classifier has operating point:

$$F = \omega F_A + (1 - \omega) F_B$$

$$T = \omega T_A + (1 - \omega) T_B$$



Scott, Martin JJ, Mahesan Niranjan, and Richard W. Prager. "Realisable Classifiers: Improving Operating Performance on Variable Cost Problems." BMVC. 1998.

Comparing classifiers

Comparing classifiers on the basis of accuracy or misclassification rate is tantamount to assuming that the costs of misclassification are equal.

The ROC gives a method to compare classifiers for all possible misclassification costs:

- ▶ **AUC** Area under the ROC curve.
 - ▶ Quantifies how closely the ROC curve approaches the top-left corner.
 - ▶ Equivalent to the Wilcoxon-Mann-Whitney statistic:
Given randomly chosen examples from class 1 and class 2, the AUC is the probability that the classifier will give a larger probability of assigning the class 1 example to class 1 (a true positive) than the class 2 example (a false positive).

Tom Fawcett. ROC Graphs: Notes and Practical Considerations for Researchers, Machine Learning, 2004.

- ▶ **Loss functions:** Trevor Hastie, Robert Tibshirani and Jerome Friedman. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. Springer, 2009.
- ▶ **Decision theory and unequal losses:** Richard Duda, Peter Hart and David Stork. *Pattern Classification*. Wiley, 2000.
- ▶ **ROC:** Tom Fawcett. *ROC Graphs: Notes and Practical Considerations for Researchers*, Machine Learning, 2004.